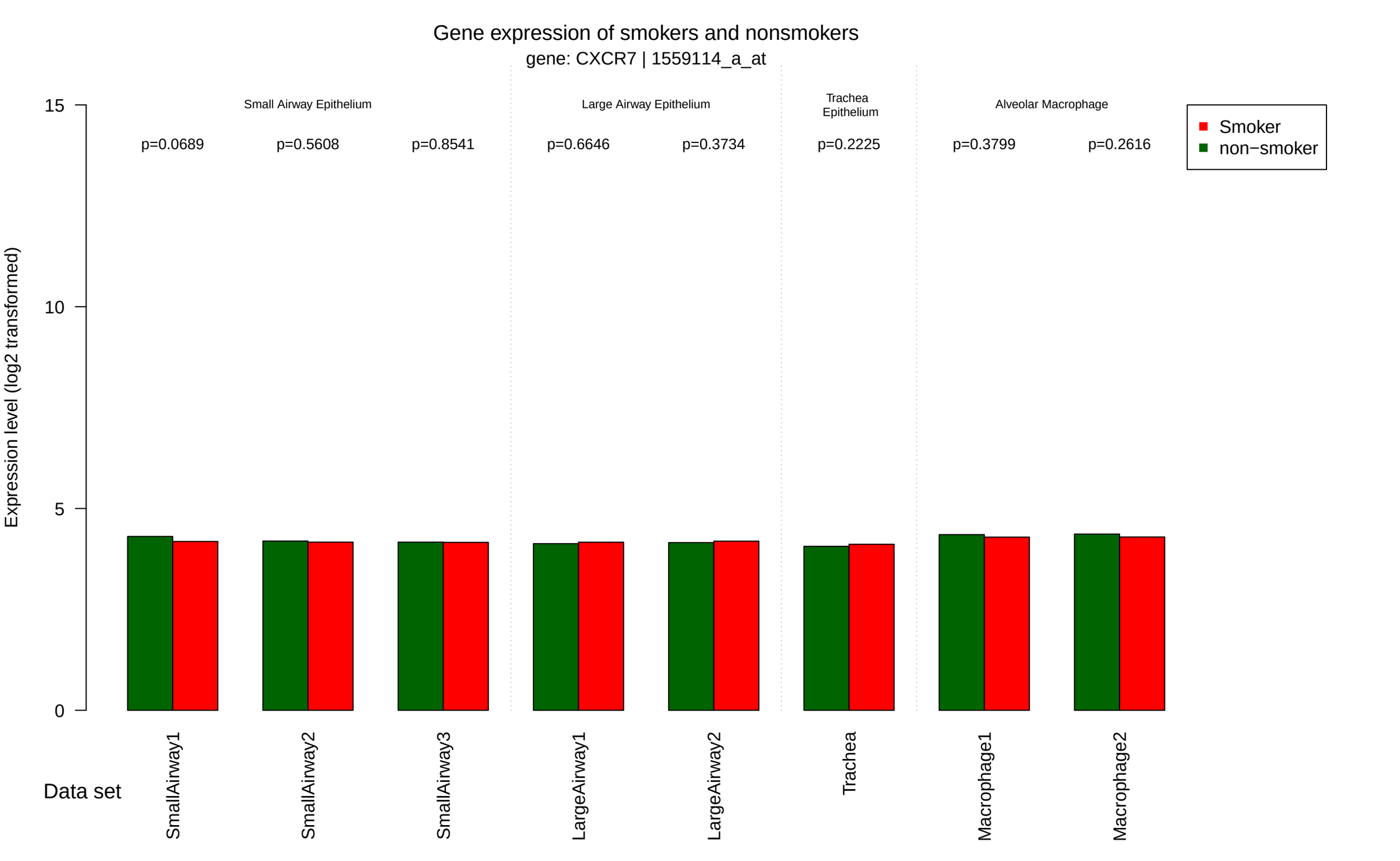
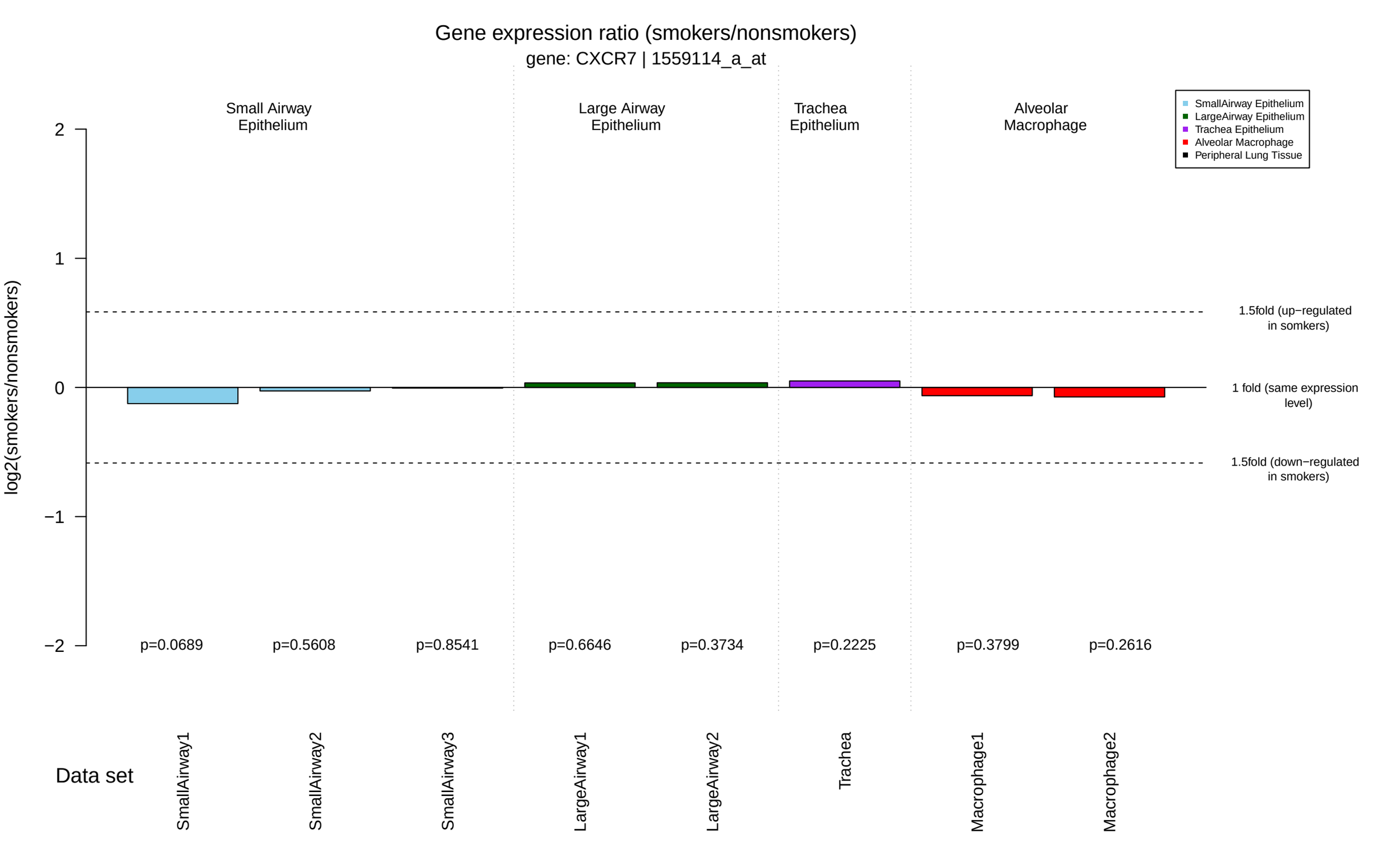
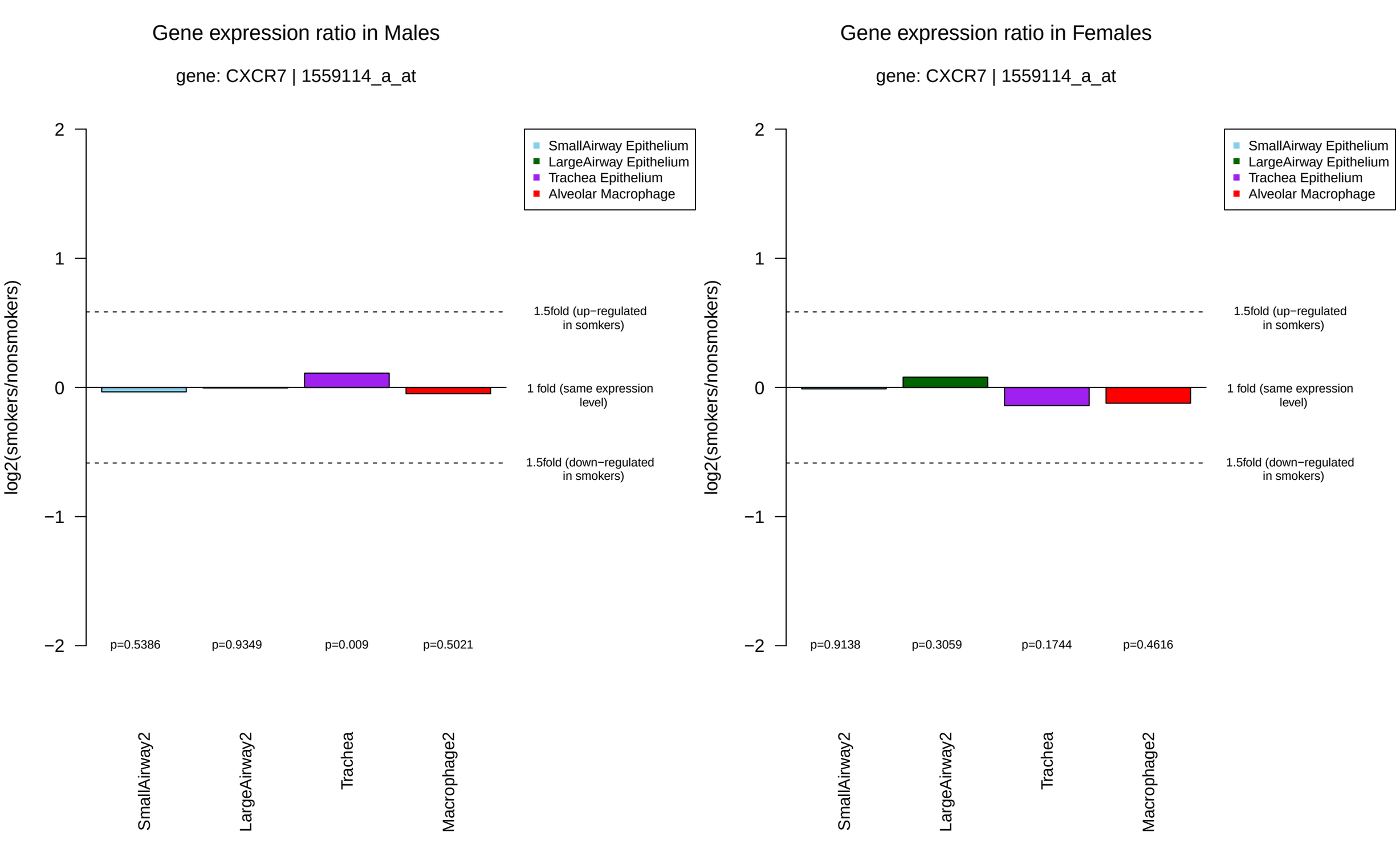
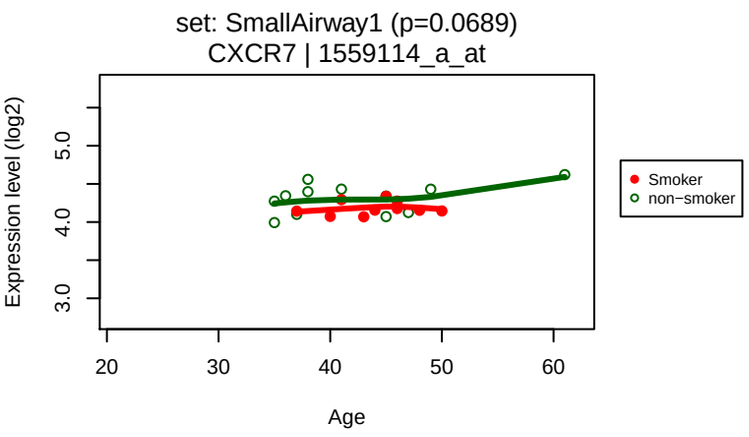
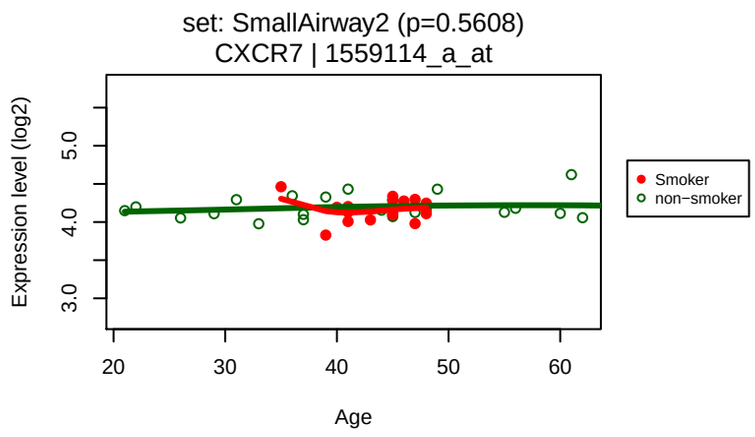
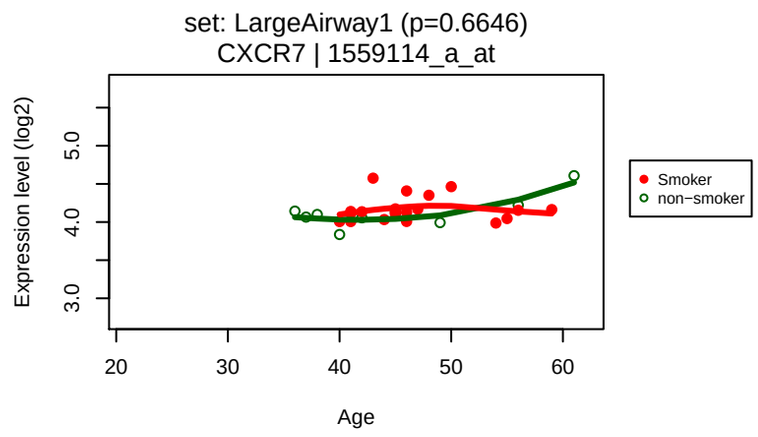

Gene expression of smokers and nonsmokers
gene: CXCR7 | 1559114_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0689
p=0.5608
p=0.8541
p=0.6646
p=0.3734
p=0.2225
p=0.3799
p=0.2616
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

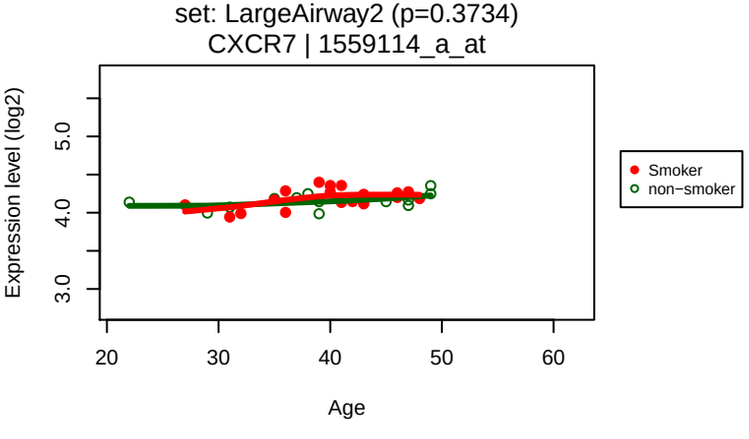

Gene expression ratio (smokers/nonsmokers)
gene: CXCR7 | 1559114_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0689
p=0.5608
p=0.8541
p=0.6646
p=0.3734
p=0.2225
p=0.3799
p=0.2616
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

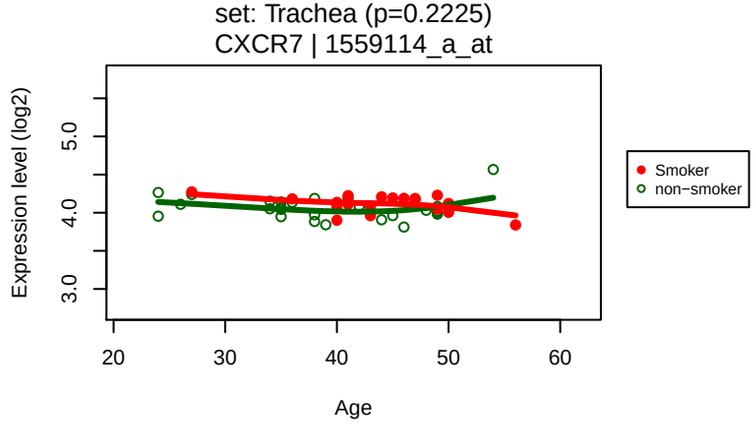

Gene expression ratio in Males
gene: CXCR7 | 1559114_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5386
p=0.9349
p=0.009
p=0.5021
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CXCR7 | 1559114_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9138
p=0.3059
p=0.1744
p=0.4616
SmallAirway2
LargeAirway2
Trachea
Macrophage2

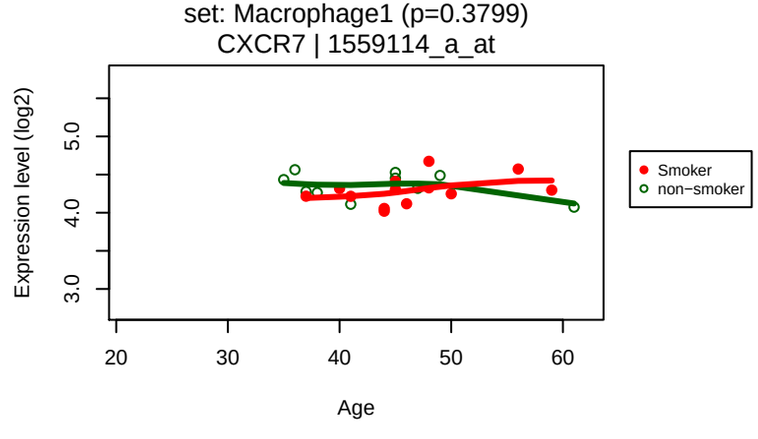

set: SmallAirway1 (p=0.0689)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

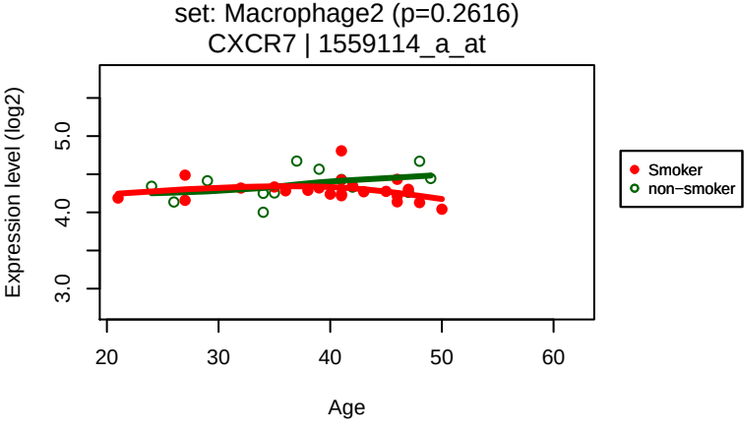

set: SmallAirway2 (p=0.5608)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

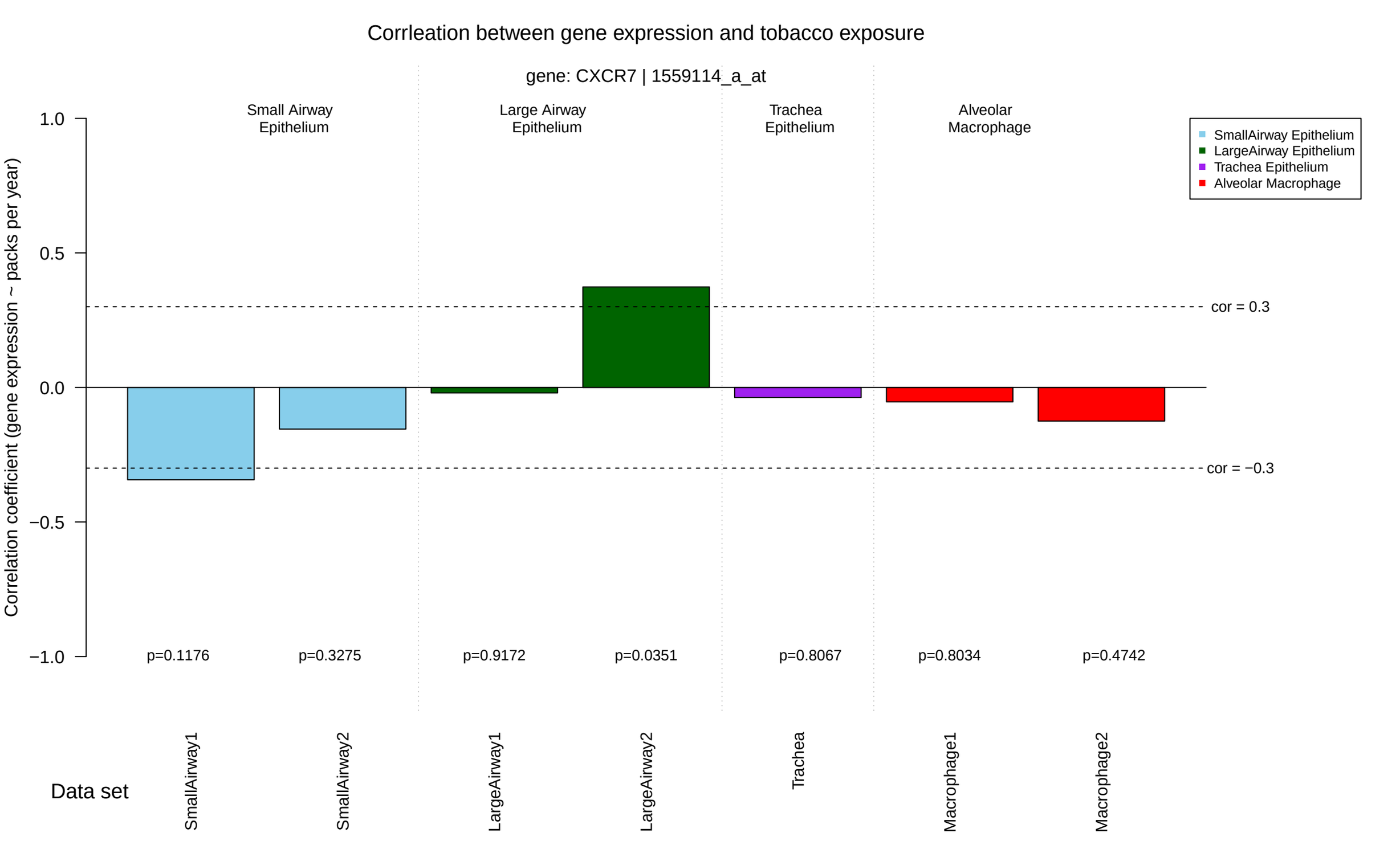

set: LargeAirway1 (p=0.6646)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3734)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2225)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3799)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2616)
CXCR7 | 1559114_a_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CXCR7 | 1559114_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1176
p=0.3275
p=0.9172
p=0.0351
p=0.8067
p=0.8034
p=0.4742
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2